

MINHAZUL ISLAM

AI Engineer · AI Infrastructure Engineer

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Summary

AI Engineer with a Master of Industrial Design Engineering from Zhejiang University and a Bachelor of Computer Science from Yunnan University. Specializes in generative AI and NLP — LLMs, RAG, prompt engineering, LangChain, and Transformer fine-tuning (RoBERTa) with PyTorch. Builds end-to-end AI systems from model development to high-concurrency production deployment with FastAPI, plus hands-on AI + hardware (Arduino, ESP32-CAM, sensor fusion, edge AI). Focus areas: multi-agent orchestration, deterministic guardrails, enterprise agent runtimes, and human-AI co-decision research.

Education

Masters in Industrial Design Engineering — Zhejiang University <i>Zhejiang, China</i>	<i>Sep 2023 – Present</i>
Bachelor in Computer Science and Technology — Yunnan University <i>Kunming, China</i>	<i>2019 – 2023</i>

Experience

AI Infrastructure & Systems Engineer (Internship) — Synexia AI
2025 – 2026 · Zhejiang, China

EDIA — Enterprise Decision Intelligence Platform

- **Multi-Agent Orchestration:** Architected a 17-node LangGraph state machine coordinating 15 specialized agents and 57 atomic skills across 5 execution tiers, with parallel branching, sequential quality gates, and multi-turn session state.
- **Deterministic Guardrail Pipeline:** Designed a pre-rendering pipeline that locks all business-critical numerical forecasts before LLM narration, with a Python validation engine enforcing exact-value compliance, language consistency, and strict source attribution — 0 hallucinated numbers reaching users.
- **Hybrid Retrieval Architecture:** Integrated ChromaDB vector search with structured SQL views, fusing unstructured market intelligence with ERP historical data — cut decision workflows from 5 minutes to under 30 seconds (SQL round-trips 77 → 2–5, rows fetched 180K → 200).
- **Production Operations:** Deployed high-concurrency SSE streaming endpoints (FastAPI, Nginx, systemd) with multi-horizon time-series forecasting (Ensemble + STL) and automated MAPE backtesting.

Zhanlu — Enterprise Multi-Agent AI Platform

- **Platform Architecture:** Solo-designed and deployed an 11-service containerized architecture (FastAPI, PostgreSQL, Redis, MinIO) with multi-tenant workspace isolation and OAuth2/JWT security.
- **Sandboxed Execution & Agent Harness Core:** Built 4 dedicated sandbox runtimes (Python / PPTX / WebApp / Office) isolating dynamic execution, plus a custom Harness agent runtime (plan → act → verify FSM) with a 100+ tool registry, per-agent allow/deny filtering, and 8 agent archetypes.
- **Dynamic Model & Tool Gateway:** Enterprise Model/Tool/MCP Gateway with hierarchical per-task model routing, provider circuit-breaker/failover, Fernet-encrypted API-key storage, native MCP client + OAuth, and complete access audit trails.

Software Automation Engineer — ProFabx
Jun 2024 – Sep 2024 · Ningbo, China

- **Parametric Pipeline Automation:** Programmed custom Python drivers using the Autodesk Inventor API (win32com.client) to automate dynamic 3D CAD modeling, cutting engineering iteration time for custom fabrications.
- **Interactive Configuration Interface:** Built a lightweight web interface (JavaScript, Python) enabling real-time parametric model updates driven directly by client specification changes.

Selected Projects

BepsBot — AI-Driven Mental-Health Peer-Support Platform (2024 – 2025)

- Built a dual-mode LLM writing assistant for bipolar-disorder peer-support communities: retrieval-augmented generation (ChromaDB + Sentence Transformers + DeepSeek LLM) with empathetic vs. cognitive-reappraisal response modes.
- Fine-tuned RoBERTa for emotion scoring; deployed high-concurrency API with FastAPI/Flask and ThreadPoolExecutor; peer-reviewed at ICEC 2025 / HHME 2026.

SPROUT — Shape-Changing Meditation Device (Sep – Dec 2023)

- Designed a shape-changing meditation device supporting academic-mindset regulation; embedded development (Arduino, sensor integration) with Fusion 360 mechanical design and 3D printing.

IoT-Based Smart Attendance System with Face Recognition (2022)

- ESP32-CAM edge face recognition (LBPH) with HTTPS IoT-cloud storage; contact-free, proxy-attendance-proof workflow validated in a live dormitory environment.

Publications

Islam, M., Xue, M., & Afra, T. (2025). Exploring Psychologist-Applied Biomarkers in Bipolar Disorder: A Systematic Framework. *ICEC 2025 IFIP TC 14 Workshops*, Tokyo, Japan, pp. 61–75. doi:10.1007/978-3-032-02534-0_8

Afra, T., Xue, M., & Islam, M. (2025). Enhancing Biofeedback Interventions for Depression and Anxiety Through Entertainment Computing: A Systematic Review. *ICEC 2025 IFIP TC 14 Workshops*, Tokyo, Japan, pp. 37–51. doi:10.1007/978-3-032-02534-0_6

Islam, M., Xue, M., & Afra, T. (2026). BepsBot: A Dual-Mode Writing Assistant for Peer Support in Bipolar Disorder Communities. *HHME 2026 — PCC Oral Presentation*, Zhejiang University, 17 Aug 2026.

Afra, T., Xue, M., & Islam, M. (2026). How Should Voice Agents Respond to Anger and Sadness? A Comparison of Empathic Response, Affect-Neutral Acknowledgment, and Cognitive Reappraisal. *HHME 2026 — PCC Oral Presentation*, Zhejiang University, 17 Aug 2026.

Certifications

- Generative AI with Large Language Models — DeepLearning.AI (Nov 2025)
- Mastering Generative AI: Fine-Tuning Transformers — IBM (Oct 2025)
- Developing Generative AI Applications with Python — IBM (Oct 2025)
- Python for AI & Development Project — IBM (Sep 2025)

Honors

- **United Nations Technology Bank — Global Youth Talent Program 2023:** Selected among top young designers worldwide; featured on the official UN Technology Bank website.

Technical Skills

Multi-Agent Systems & Agent Runtimes: LangGraph (17-node FSM), Agent Harness, Model Context Protocol (MCP), plan-first turn planning, sub-agent workflows, goal contracts, LLM-as-a-judge evaluation, Hermes control layer

Context Engineering & Retrieval: SafeContextGate budget enforcement, hybrid vector + SQL RAG, ChromaDB / BAAI/bge-m3 embeddings, deterministic guardrails, output validation, context trimming & isolation

AI Infrastructure & LLMops: Docker stacks (14 services), FastAPI / SSE streaming, local vLLM (Qwen3-72B) / Ollama, Redis, PostgreSQL / MinIO, Prometheus & Grafana, multi-provider model routing, dual-LLM security gates, systemd / Nginx / auto-recovery

Languages & AI Development: Python (asyncio), C/C++, SQL, JavaScript, PyTorch / Hugging Face Transformers, PEFT/LoRA fine-tuning, Git / GitHub Actions CI/CD

Edge AI & Embedded HCI: Arduino, ESP32-CAM, sensor fusion, edge AI prototyping, Fusion 360 / Autodesk Inventor API automation